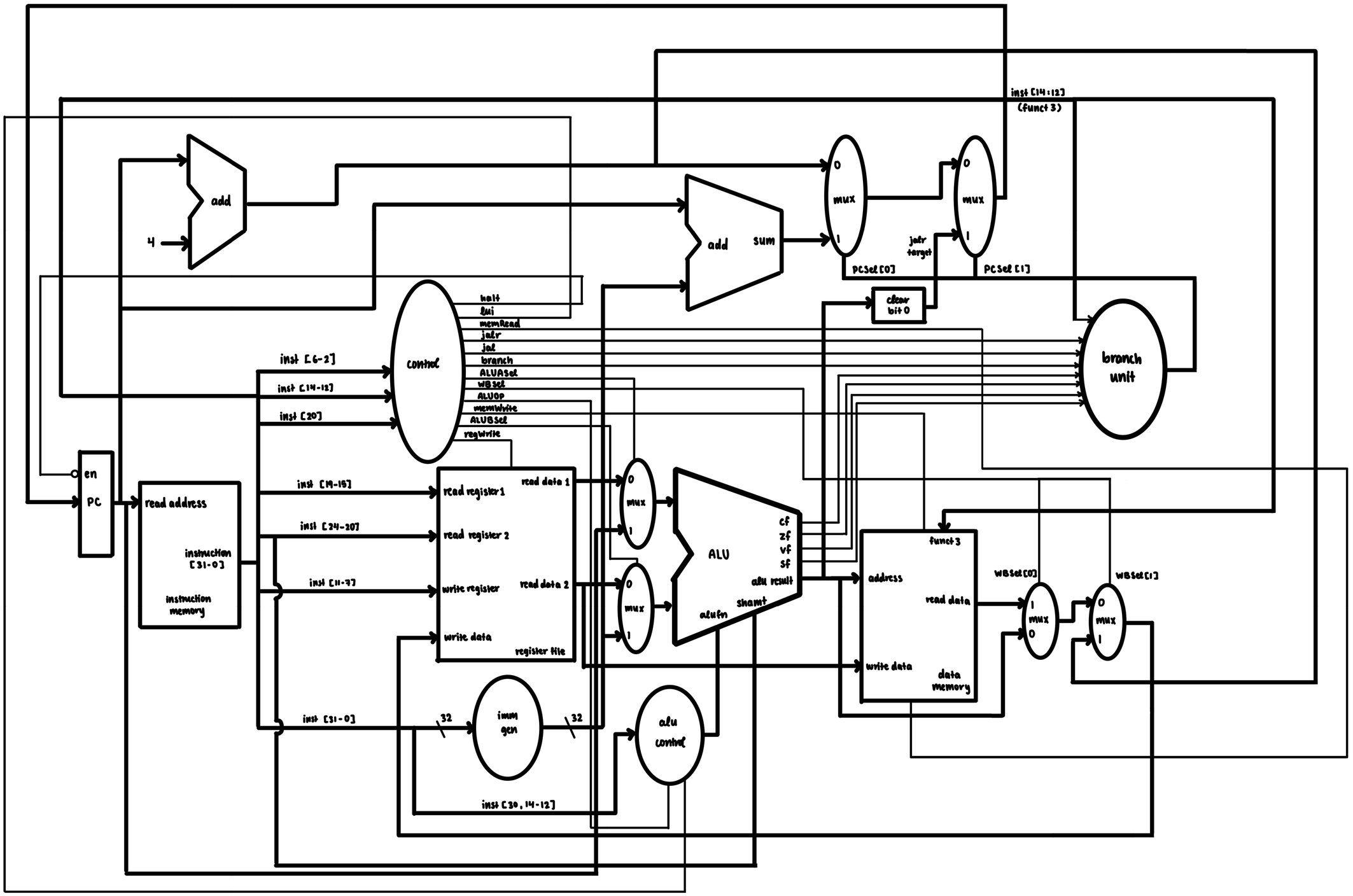

add
4
add
sum
0
mux
1
PCSel [0]
0
mux
1
PCSel [1]
jalr target
inst [14:12]
(funct 3)
clear bit 0
branch unit
halt
lui
memRead
jalr
jal
branch
ALUASel
WBSel
ALUOp
memWrite
ALUBSel
regWrite
control
inst [6-2]
inst [14-12]
inst [20]
en
PC
read address
instruction [31-0]
instruction memory
inst [19-15]
inst [24-20]
inst [11-7]
read register 1
read register 2
write register
write data
read data 1
read data 2
register file
0
mux
1
0
mux
1
ALU
cf
zf
vf
sf
alu result
shamt
alufn
funct3
address
read data
write data
data memory
WBSel [0]
WBSel [1]
1
mux
0
0
mux
1
inst [31-0]
32
imm gen
32
alu control
inst [30, 14-12]